

Machine Learning

End-Semester Examination

Duration: 3 hours

Date: May 12' 2023

Marks: 50

Please outline all the steps systematically in the examination booklet.

1. (a) Suppose that we are given a sequence of training samples $\{x_1, x_2, \dots, x_N\}$, that are generated from a univariate normal distribution $N(\mu, 1)$. It is known that our prior knowledge on the mean μ follows a normal distribution $N(3, 2)$. Compute the MAP estimate of the mean μ . The normal distribution $N(\mu, \sigma^2)$ is given by

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$$\frac{3 + 2\bar{x}}{1 + 2N}$$

[4 marks]

2. Consider a category ω_1 having a Geometric distribution $p(1-p)^{x-1}$, $0 \leq p \leq 1$ as the class conditional density, with parameter p . Note that x is discrete random variable. Given the set of N one dimensional training data $\{x_1, x_2, \dots, x_N\}$, estimate the ML for the parameter p .

[4 marks]

3. A one-dimensional Gaussian Mixture Model, (with **known** values of identical variances $\sigma^2 = 1$ and mixing coefficients = $\frac{1}{3}$ and $\frac{2}{3}$) is defined below:

$$p(x|\omega) = \frac{1}{3} \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2} \right) + \frac{2}{3} \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-4)^2} \right)$$

- (i) If a data point is to be generated from this GMM, what will the probability of choosing the first Gaussian density term, prior to generation.
- (ii) Compute the responsibilities of the individual Gaussians for a sample point $x = 2$.

[2 + 4 marks]

4. A multi layer perceptron is trained to recognize the digits in $S = \{1, 2, 3\}$ by using a batch of samples. The targets are encoded using the one-hot scheme, whereby the i^{th} neuron in the output layer can provide the probability for the i^{th} digit in S . A simple stochastic gradient descent scheme is adopted for back-propagation

- (a) If the feature vector representing the digit 2 is sent as input, what will the target vector $\mathbf{t}$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{matrix} 0.6914 \\ 0.3086 \end{matrix}$$

- (b) If the predicted values at the output neurons are (2, 6, 8) respectively, use softmax function to map them to probability values. $[0.0021785 \quad 0.118993 \quad 0.8788285]$
- (c) By considering the targets and mapped probability values in (a) and (b), compute the
- multi-category cross entropy loss 2.129110892
 - mean squared loss 1.5486977

[1 + 3 + 4 = 8 marks]

- ✓ 5. In the training of a network, the following 4×4 patch in a feature map is analysed

$$X = \begin{bmatrix} 1 & 3 & 2 & -8 \\ 1 & 3 & -4 & -7 \\ -1 & \mathbf{7} & -1 & 3 \\ -2 & 3 & 4 & 9 \end{bmatrix}$$

Compute the output when,

- (a) ReLU activation is applied to the coefficients of X .
- (b) Sigmoid function is applied to the pixel marked in **bold** in X

$$0.999052948$$

[2 + 2 = 4 marks]

6. You are given 3 training examples (comprising 2 dimensions $[x_1 x_2]$) for classes ω_1 and ω_2 with labels $y_1 = 1$ and $y_2 = -1$. A single layer perceptron is used for the learning.

$$\omega_1: \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \omega_2: \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$\underset{1}{\quad} \quad \underset{-3}{\quad} \quad \underset{3}{\quad} \quad \quad \underset{-3}{\quad} \quad \underset{-5}{\quad} \quad \underset{1}{\quad}$

- (a) At a given iteration, if $x_1 + x_2 - 2 = 0$ is used as a decision boundary, specify the misclassified samples (if any). Clearly state their corresponding value of the discriminant function.

- (b) Recall that, in a perceptron learning, we update the weights by accumulation of sample feature vectors being misclassified. Use this knowledge to obtain the decision boundary for the subsequent iteration

$$-2x_1 - 2 = 0$$

[3 + 3 = 6 marks]

7. In a two class, two dimensional problem, the feature vectors in each class are normally distributed with identical covariance matrix: $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$. The respective mean vectors μ_1 and μ_2 (for class ω_1 and ω_2) are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Assuming that the classes ω_1 and

ω_2 are probable with 0.6 and 0.4, classify the feature vector $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$, according to the Bayes minimum probability classifier. [4 marks]

The normal distribution $N(\mu, \Sigma)$ in d -dimensions is given by

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{0.5}} e^{-\frac{1}{2}(\mathbf{x}-\mu)^T \Sigma^{-1}(\mathbf{x}-\mu)} \quad \omega_1$$

8. Consider a multi-layer perceptron with the following specifications:

- d nodes in the input layer
- sigmoidal activation functions at the M nodes in the hidden layer.
- linear activation functions at the c nodes in the output layer.

The weights of the network are to be learnt with N training samples $\{(\mathbf{x}_i, t_i)\}_{i=1}^N$ using the error function

$$E = \sum_{n=1}^N E^n = \frac{1}{6} \sum_{n=1}^N \sum_{k=1}^c (t_k^n - y_k^n)^6$$

Each of the training samples comprise d dimensions. Derive expressions for the following:

- Predicted output $y_k^n = y_k(\mathbf{x}_n, \mathbf{w})$ in terms of the input $\mathbf{x}_n$, activation functions and network weights.
- Updation of weight w_{kj} , connecting the j^{th} hidden node to the k^{th} output node.
- Updation of weight w_{ji} , connecting the i^{th} input node to the j^{th} hidden node. Note: For the sigmoid $g(x) = \frac{1}{1+e^{-x}}$, the derivative is $g(x)(1-g(x))$.

[4 + 4 + 4 = 12 marks]

9. Assume that we have a single training pair $(\mathbf{x}_1, y_1)$, where $\mathbf{x}_1 \neq 0$. If we used the LMS Algorithm to adjust the weights, defined by

$$\mathbf{w}_{k+1} = \mathbf{w}_k + \eta \frac{e_k \mathbf{x}_1}{\mathbf{x}_1^T \mathbf{x}_1}$$

where

$$e_k = (y_1 - \mathbf{w}_k^T \mathbf{x}_1)$$

Find an expression for e_{k+1} as a function of e_k and η .

[2 marks]

$$(1 - \eta) e_k$$